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1 System Modelling and Problem Formulation

In this chapter, the multi-prosumer building energy management problem investigated in this thesis is defined, and the physical models and mathematical formulations involved in the environment are presented. The system under study consists of three prosumers connected to a low-voltage distribution network. Each prosumer is equipped with local electrical loads, photovoltaic (PV) generation, a stationary battery energy storage system (BESS), and residential electric vehicle (EV) charging demand. At each discrete time step, the control system determines the battery charging/discharging power, PV utilization, and EV charging power to minimize the total electricity purchasing cost while satisfying both the physical constraints of the devices and the operational safety constraints of the distribution network.

This chapter focuses only on the system modeling and problem formulation, including the system constraints and the optimization objective function. The details of the control methodology, such as how reinforcement learning actions are generated, how the EV control strategies are implemented at the control layer, and how the grid safety projection modifies the control actions, are discussed in Chapter 4.

1.1 Overall System

The system investigated in this thesis addresses the energy management problem of a low-voltage distribution network integrated with distributed energy resources. The system consists of a low-voltage distribution network, 3 prosumer agents, household loads, PV generation, a stationary battery energy storage system, EV charging, electricity price signals, and a power flow calculation module. Each prosumer acts not only as an electricity consumer but also as an electricity producer and the owner of dispatchable energy resources. The primary objective of the system is to coordinate the resources of the three prosumers to minimize the overall operating cost while satisfying both the device operational constraints and the distribution network safety constraints.

The distribution network topology is adopted from SimBench, with the fixed network configuration

1-LV-rural1--0-sw.

The system consists of $N = 3$ prosumers connected to the distribution network buses

$$\mathcal{B} = \{12, 4, 2\}.$$

Let $i \in \mathcal{I} = \{1, 2, 3\}$ denote the prosumer index, and $t \in \mathcal{T} = \{0, \dots, T - 1\}$ represent the discrete time step within an episode. The simulation uses a 15-minute resolution, so a day can be represented as

$$\Delta t = 0.25 \text{ h}, \quad T = 96.$$

One episode corresponds to the operation of a single day. During training, multiple consecutive days can be concatenated to form a longer training window.

The local power profile of each prosumer i consists of external inputs and three kinds of controllable or semi-controllable energy resources. The external inputs include the household load $L_{i,t}^{hh}$, the available PV generation $G_{i,t}$, and the electricity price p_t . In this thesis, only the household load is considered as the base electrical demand. The heat pump load is not included in the current model but can be incorporated as an additional load in future work:

$$L_{i,t} = L_{i,t}^{hh}.$$

The controllable resources include the charging and discharging power of the stationary battery $P_{i,t}^b$, the utilized PV generation $G_{i,t}^{use}$, and the EV charging power $P_{i,t}^{ev}$. Accordingly, the electricity-related variables of each prosumer can be expressed as

$$\{L_{i,t}, G_{i,t}, P_{i,t}^b, P_{i,t}^{ev}\},$$

where $L_{i,t}$ denotes the total local electrical load, $G_{i,t}$ denotes the available PV generation, $P_{i,t}^b$ denotes the stationary battery power, and $P_{i,t}^{ev}$ denotes the EV charging power. In this thesis, the following sign convention is adopted for the battery power: $P_{i,t}^b > 0$ indicates battery charging, whereas $P_{i,t}^b < 0$ indicates battery discharging.

$$-P_i^{b,\max} \leq P_{i,t}^b \leq P_i^{b,\max}.$$

Since only EV charging is considered in this thesis, while vehicle-to-grid is not allowed, the EV charging power satisfies

$$P_{i,t}^{ev} \geq 0.$$

At each time step, the power of all devices within a prosumer is aggregated to determine the net load injected into the distribution network:

$$P_{i,t}^{net} = L_{i,t} - G_{i,t}^{use} + P_{i,t}^b + P_{i,t}^{ev}.$$

When $P_{i,t}^{net} > 0$, the prosumer imports electricity from the distribution grid; when $P_{i,t}^{net} < 0$, the prosumer exports power back to the grid. The net loads of the three prosumers are injected into the low-voltage distribution network model, where power flow analysis is performed using pandapower. The power flow calculation returns the bus voltages, line loading ratios, and transformer loading ratios, which are subsequently used to determine whether the control actions satisfy the distribution network safety constraints.

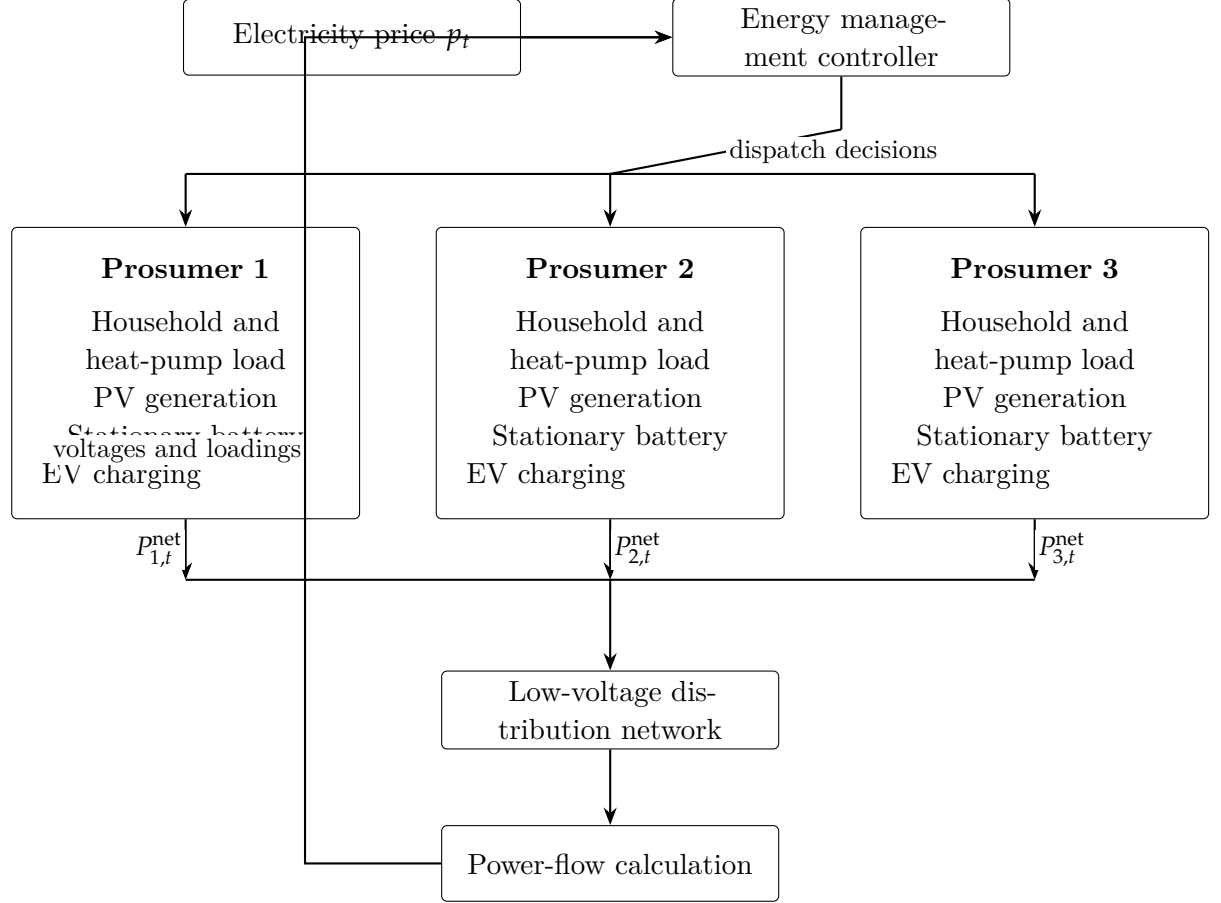


Figure 1.1: Overall architecture of the system investigated in this thesis.

Figure 1.1 illustrates the overall architecture of the system. The household load, heat pump load, PV generation, and electricity price are provided as time-series inputs from the dataset. The stationary battery and EV charging serve as the dispatchable resources on the prosumer side. The grid power flow calculation module maps the net load of each prosumer to the corresponding network operating states, including bus voltages and equipment loading conditions. Accordingly, the problem addressed in this thesis can be

formulated as follows: at each discrete time step, the control system determines $P_{i,t}^b$, $G_{i,t}^{use}$, and $P_{i,t}^{ev}$ for each prosumer based on the load demand, PV generation, electricity price, and the state of charge of the energy storage devices, with the objective of minimizing the total operating cost while ensuring the safe operation of the stationary battery, the EV, and the distribution network.

1.2 Load of the Prosumers

The base electrical load of each prosumer is represented by the household load obtained from the original dataset. The corresponding data file is located at

`datasets/prosumer/household.csv`.

The load profiles of the three prosumers are

SFH12, SFH18, SFH20.

Here, SFH denotes the load profile of a single-family house. The original dataset is read using UTC timestamps and subsequently converted to the Europe/Berlin time zone to ensure that the EV arrival and departure times are consistent with the local time.

Let $L_{i,t}^{hh,raw}$ denote the original household load. In the implementation, an agent-specific scaling factor s_i^L is applied to adjust the original load to the scale used in this study. The default setting is

$$s_1^L = s_2^L = s_3^L = 20.$$

Accordingly, the total electrical load of prosumer i at time step t is given by

$$L_{i,t} = s_i^L \left(L_{i,t}^{hh,raw} \right).$$

This load represents only the external electricity demand of the prosumer and does not include PV generation, battery charging/discharging, or EV charging. These components are incorporated later.

1.3 PV Modelling

The available PV power is obtained from the data file

`datasets/prosumer/pv_reference.csv`.

In the current configuration, the reference column `ref_south` is used. The same reference PV profile is assigned to all three prosumers and scaled by an agent-specific scaling factor s_i^G ,

$$s_1^G = s_2^G = s_3^G = 1.$$

Accordingly, the available PV generation is given by

$$G_{i,t} = s_i^G G_t^{ref}.$$

The controller does not directly modify the available PV generation $G_{i,t}$; instead, it determines the utilized PV power $G_{i,t}^{use}$. Let $u_{i,t}^{pv} \in [0, 1]$ denote the PV utilization factor. Then,

$$\begin{aligned} G_{i,t}^{use} &= u_{i,t}^{pv} G_{i,t}, \\ G_{i,t}^{curt} &= G_{i,t} - G_{i,t}^{use}. \end{aligned}$$

$G_{i,t}^{curt}$ denotes the curtailed PV power. In the current model, no explicit penalty is assigned to PV curtailment. Therefore, PV curtailment primarily affects the system performance indirectly by changing the net load and the operating state of the distribution network.

1.4 Stationary Battery Energy Storage Model

Each prosumer is equipped with a stationary battery. Let C_i^b denote the battery capacity, and let $SOC_{i,t}^b$ denote the battery state of charge (SoC) at the beginning of time step t . The default parameters are

$$C_i^b = 100 \text{ kWh}, \quad SOC_{\min}^b = 0.05, \quad SOC_{\max}^b = 0.95.$$

The battery charging and discharging efficiency is

$$\eta_b = 0.95.$$

Maximum charging and discharging powers are determined by the capacity and maximum charging rate:

$$P_i^{b,\max} = r_{\max}^b C_i^b,$$

where the default $r_{\max}^b = 0.5$, thus a 100 kWh battery corresponds to

$$P_i^{b,\max} = 50 \text{ kW}.$$

The battery power constraints are

$$-P_i^{b,\max} \leq P_{i,t}^b \leq P_i^{b,\max}.$$

The battery SoC dynamics are determined by the charging and discharging directions as mentioned before. If $P_{i,t}^b \geq 0$, the battery is in charging state; if $P_{i,t}^b < 0$, the battery is in discharging state. The SoC is updated as

$$SOC_{i,t+1}^b = \begin{cases} SOC_{i,t}^b + \frac{P_{i,t}^b \Delta t \eta_b}{C_i^b}, & P_{i,t}^b \geq 0, \\ SOC_{i,t}^b + \frac{P_{i,t}^b \Delta t}{\eta_b C_i^b}, & P_{i,t}^b < 0. \end{cases}$$

subject to the boundary constraint

$$SOC_{\min}^b \leq SOC_{i,t+1}^b \leq SOC_{\max}^b.$$

This formulation captures the effect of charging and discharging efficiency losses. During charging, only a fraction η_b of the input electrical energy is stored in the battery. During discharging, a larger amount of internal battery energy must be consumed to deliver the same amount of electrical energy to the external system.

1.5 EV Modelling

Each prosumer is equipped with one residential EV. In the distribution network model, the EV is represented as a controllable charging load characterized by a connection time window, a charging power limit, a battery capacity, and a required departure SoC. In this thesis, vehicle-to-grid (V2G) operation is not considered. Therefore, the EV cannot discharge power to either the building or the distribution network and can only draw electrical energy while connected to the residential charger. Let C_i^{ev} denote the EV battery capacity, and let $SOC_{i,t}^{ev}$ denote the state of charge (SoC) of the EV at the beginning of time step t . The default parameters are

$$C_i^{ev} = 60 \text{ kWh}, \quad SOC_{\min}^{ev} = 0.10, \quad SOC_{\max}^{ev} = 0.95.$$

The maximum power of the residential charger is

$$P_i^{ev,\max} = 11 \text{ kW},$$

and the EV charging efficiency is

$$\eta_{ev} = 0.95.$$

Since the simulation time step is $\Delta t = 0.25 \text{ h}$, the maximum amount of energy that can be charged during a single 15-minute time step is

$$E_{i,\Delta t}^{ev,\max} = P_i^{ev,\max} \Delta t \eta_{ev}.$$

1.5.1 EV Connection Window

EV's connection status is determined by the intraday time steps. The current assumption is that the vehicle arrives home at 18:00 and connects to the residential charger, and departs at 07:00 the next day. Since there are 96 fifteen-minute time steps in a day, the arrival and departure times are

$$t^{arr} = 72, \quad t^{dep} = 28.$$

Let $t_d = t \bmod T$ denote the intraday time step. Since $t^{arr} > t^{dep}$, the connection window spans midnight, and thus a complete parking window consists of the time steps $72, \dots, 95$ from the evening of the current day and the time steps $0, \dots, 27$ from the early morning of the next day:

$$\mathcal{T}^{ev} = \{t : t_d \geq t^{arr}\} \cup \{t : t_d < t^{dep}\}.$$

The connecting status of the EV is represented by the variable A_t^{ev} :

$$A_t^{ev} = \begin{cases} 1, & t_d \geq t^{arr} \text{ or } t_d < t^{dep}, \\ 0, & \text{otherwise.} \end{cases}$$

When $A_t^{ev} = 1$, the vehicle is at home and can be charged; when $A_t^{ev} = 0$, the vehicle is away, and the residential charging power must be zero.

1.5.2 EV 到家 SoC 与充电功率约束

Upon arrival at home, the EV SoC is initialized to the arrival SoC:

$$SOC_{i,t}^{ev} = SOC_{arr}^{ev} = 0.30, \quad t_d = t^{arr}.$$

Thereafter, until the EV departs, its SoC can only increase through residential charging. The EV charging power must satisfy both the connection status and the charger power limit:

$$0 \leq P_{i,t}^{ev} \leq A_t^{ev} P_i^{ev, \max}.$$

Furthermore, the EV SoC must not exceed its upper limit. Expressing the remaining SoC headroom as an equivalent charging power yields an additional upper bound determined by the remaining battery capacity:

$$P_{i,t}^{ev} \leq \frac{(SOC_{\max}^{ev} - SOC_{i,t}^{ev})C_i^{ev}}{\Delta t \eta_{ev}}.$$

Therefore, the physically feasible upper bound on the EV charging power can be written as

$$0 \leq P_{i,t}^{ev} \leq \min \left(A_t^{ev} P_i^{ev, \max}, \frac{(SOC_{\max}^{ev} - SOC_{i,t}^{ev})C_i^{ev}}{\Delta t \eta_{ev}} \right).$$

When the EV is not connected, i.e., $A_t^{ev} = 0$, the above formulation automatically yields $P_{i,t}^{ev} = 0$. As the EV approaches $SOC_{\max}^{ev}$, the SoC headroom constraint progressively limits the charging power, thereby preventing the battery from being charged beyond its physical capacity.

1.5.3 EV SoC Status

The change of the EV SoC is determined by the actual charging power:

$$SOC_{i,t+1}^{ev} = SOC_{i,t}^{ev} + \frac{P_{i,t}^{ev} \Delta t \eta_{ev}}{C_i^{ev}}.$$

Since vehicle-to-grid operation is not considered in this thesis, $P_{i,t}^{ev} \geq 0$. Consequently, the EV SoC is monotonically non-decreasing throughout the connection window. During time steps when the EV is not connected and no arrival SoC reset occurs, the EV does not exchange energy with the residential charging system. At all time steps, the EV SoC must satisfy the following physical constraints:

$$SOC_{\min}^{ev} \leq SOC_{i,t}^{ev} \leq SOC_{\max}^{ev}.$$

1.5.4 EV Departure SoC Requirement

Another physical requirement for the EV is to have sufficient charge when departing. The default departure target SoC is set to

$$SOC_{req}^{ev} = 0.90.$$

Therefore, for each prosumer i , the SoC requirement at departure time is

$$SOC_{i,t^{dep}}^{ev} \geq SOC_{req}^{ev}.$$

Since the connection window spans midnight, this constraint links the charging decisions made after the EV arrives home in the evening to its availability for use on the following morning. Equivalently, the minimum increase in battery energy required to charge the EV from the arrival SoC to the required departure SoC is

$$\Delta E_i^{ev,req} = \max(0, SOC_{req}^{ev} - SOC_{arr}^{ev}) C_i^{ev}.$$

Considering the charging efficiency, the minimum amount of energy that must be supplied from the grid is

$$E_i^{ev,grid,req} = \frac{\Delta E_i^{ev,req}}{\eta_{ev}}.$$

Under the default parameter settings, charging the EV from 0.30 to 0.90 requires an increase of $0.60 \times 60 = 36$ kWh in the battery energy. Accounting for the charging efficiency $\eta_{ev} = 0.95$, the corresponding energy that must be drawn from the grid is approximately

$$\frac{36}{0.95} \approx 37.9 \text{ kWh}.$$

1.6 Electricity Price Model

The electricity price data are obtained from

`datasets/prosumer/price.csv`.

Let p_t denote the wholesale electricity price, expressed in EUR/kWh. The electricity import price used in both the environment and the optimizer is defined as

$$\tilde{p}_t = p_t + \mu^{imp},$$

where μ^{imp} denotes the electricity import price markup. In the current default configuration,

$$\mu^{imp} = 0.$$

Therefore, under the default setting $\tilde{p}_t = p_t$

1.7 Net Load, Power Flow, and Grid Safety

After integrating the load, PV, battery, and EV contributions, the net load injected by prosumer i into the distribution grid at time step t is given by

$$P_{i,t}^{net} = L_{i,t} - G_{i,t}^{use} + P_{i,t}^b + P_{i,t}^{ev}.$$

As mentioned before, when $P_{i,t}^{net} \geq 0$, the prosumer draws power from the grid; when $P_{i,t}^{net} < 0$, it injects power into the grid.

In the pandapower power flow model, positive net load is written to the load table, and negative net load is written to the sgen table:

$$P_{i,t}^{load} = \frac{\max(P_{i,t}^{net}, 0)}{1000},$$

$$P_{i,t}^{sgen} = \frac{\max(-P_{i,t}^{net}, 0)}{1000}.$$

The factor 1000 is used to convert kW to MW. Subsequently, the environment performs AC power flow calculations to obtain the voltages at each bus, line loading rates, and transformer loading rates. In other words, grid safety is not determined by simple algebraic power balance but is jointly decided by the distribution grid topology, line impedances, transformer parameters, and the net injections from each prosumer.

1.7.1 Voltage Constraints

Let $V_{b,t}$ denotes the voltage magnitude at bus b at time step t , expressed in per unit. The voltage safety range is given by

$$V_{\min} = 0.95, \quad V_{\max} = 1.05.$$

Therefore, all buses should satisfy

$$V_{\min} \leq V_{b,t} \leq V_{\max}, \quad b \in \mathcal{B}^{grid}.$$

For any bus b , the voltage violation can be written as

$$v_{b,t} = \max(0, V_{\min} - V_{b,t}) + \max(0, V_{b,t} - V_{\max}).$$

The total voltage violation across the network is measured as

$$\psi_t^V = \sum_{b \in \mathcal{B}^{grid}} v_{b,t}^2.$$

When all bus voltages are within the range $[0.95, 1.05]$, $v_{b,t} = 0$ and $\psi_t^V = 0$. If certain buses fall below or exceed the safety limits due to high load, centralized EV charging, or high PV generation, then ψ_t^V will become positive.

1.7.2 Line Loading Constraints

The safety of lines is measured by their loading percentage. Let $\lambda_{\ell,t}$ denote the loading percentage of line ℓ at time step t . Physically, $\lambda_{\ell,t}$ represents the ratio of the line's apparent power or current to its rated capacity. The current safety limit is

$$\lambda_{\max}^{line} = 100.$$

Therefore, all lines should satisfy

$$\lambda_{\ell,t} \leq \lambda_{\max}^{line}, \quad \ell \in \mathcal{L}.$$

The line violation measure is expressed as

$$\psi_t^{line} = \sum_{\ell} \left(\frac{\max(0, \lambda_{\ell,t} - \lambda_{\max}^{line})}{100} \right)^2,$$

where dividing by 100 is to convert the percentage violation into a relative measure. When all line loadings do not exceed 100%, $\psi_t^{line} = 0$.

1.7.3 Transformer Loading Constraints

Transformer safety is similarly measured by the loading percentage. Let $\lambda_{m,t}^{tr}$ denote the loading percentage of transformer m at time step t . The current safety limit for transformer loading is

$$\lambda_{\max}^{tr} = 100.$$

Therefore, all transformers should satisfy

$$\lambda_{m,t}^{tr} \leq \lambda_{\max}^{tr}, \quad m \in \mathcal{M}^{tr}.$$

The corresponding transformer violation measure is

$$\psi_t^{tr} = \sum_m \left(\frac{\max(0, \lambda_{m,t}^{tr} - \lambda_{\max}^{tr})}{100} \right)^2.$$

Transformer constraints are particularly sensitive to the system's total net load. When multiple prosumers charge their EVs or batteries simultaneously, the power demand at the feeder head may increase, thereby raising transformer loading; when PV generation is high and local load is low, reverse power flow may also affect transformer and line states.

In summary, grid safety constraints can be summarized as

$$\begin{aligned} V_{\min} &\leq V_{b,t} \leq V_{\max}, & b &\in \mathcal{B}^{grid}, \\ \lambda_{\ell,t} &\leq \lambda_{\max}^{line}, & \ell &\in \mathcal{L}, \\ \lambda_{m,t}^{tr} &\leq \lambda_{\max}^{tr}, & m &\in \mathcal{M}^{tr}. \end{aligned}$$

These safety metrics do not alter the physical power flow results themselves, but are used to evaluate whether given control actions lead to voltage violations, line overloads, or transformer overloads.

1.8 Optimization Objective

The objective of this study is to minimize the total electricity cost of the system while satisfying both the device operational constraints and the distribution network constraints. For prosumer i , the economic cost associated with the stationary battery at time step t is given by

$$C_{i,t}^b = \tilde{p}_t \Delta t P_{i,t}^b.$$

Since $P_{i,t}^b > 0$ denotes battery charging, $C_{i,t}^b$ represents a positive charging cost. Conversely, when $P_{i,t}^b < 0$, the battery is discharging, resulting in a negative value of $C_{i,t}^b$,

which corresponds to a reduction in the electricity purchasing cost or, equivalently, revenue from battery discharge.

The EV charging cost is given by

$$C_{i,t}^{ev} = \tilde{p}_t \Delta t P_{i,t}^{ev}.$$

In the current model, PV curtailment does not have an explicit economic price, thus

$$C_{i,t}^{pv} = 0.$$

Therefore, the total economic cost of the single-step system is

$$C_t = \sum_{i=1}^N \left(C_{i,t}^b + C_{i,t}^{ev} + C_{i,t}^{pv} \right).$$

In an episode, the objective function can be formulated as

$$\begin{aligned} \min_{\{P^b, G^{use}, P^{ev}\}} \quad & \sum_{t=0}^{T-1} \sum_{i=1}^N \tilde{p}_t \Delta t \left(P_{i,t}^b + P_{i,t}^{ev} \right) \\ \text{s.t.} \quad & L_{i,t} = s_i^L \left(L_{i,t}^{hh,raw} + L_{i,t}^{hp,raw} \right), \\ & 0 \leq G_{i,t}^{use} \leq G_{i,t}, \\ & -P_i^{b,max} \leq P_{i,t}^b \leq P_i^{b,max}, \\ & SOC_{min}^b \leq SOC_{i,t}^b \leq SOC_{max}^b, \\ & 0 \leq P_{i,t}^{ev} \leq A_i^{ev} P_i^{ev,max}, \\ & SOC_{min}^{ev} \leq SOC_{i,t}^{ev} \leq SOC_{max}^{ev}, \\ & SOC_{i,t}^{ev,dep} \geq SOC_{req}^{ev}, \\ & P_{i,t}^{net} = L_{i,t} - G_{i,t}^{use} + P_{i,t}^b + P_{i,t}^{ev}, \\ & V_{min} \leq V_{b,t} \leq V_{max}, \\ & \lambda_{\ell,t} \leq \lambda_{max}^{line}, \\ & \lambda_{m,t}^{tr} \leq \lambda_{max}^{tr}. \end{aligned}$$

The problem has a multi-time-step coupling characteristic: the battery SoC and EV SoC make the current actions affect the future feasible region; PV, load, and price exhibit obvious time variations; and the distribution network flow causes different prosumer actions to be coupled through voltage and line loading. Therefore, this study will model the system as a multi-agent decision problem and compare the performance of MADRL, MPC, and baselines in terms of economic efficiency and grid safety.

2 Methodology

This chapter presents the prediction, control, safety correction, and training methods adopted in this thesis. Building upon the system model introduced in Chapter 3, this chapter further describes how these models are implemented in the code. The overall framework consists of an LSTM-based forecasting module, a pandapower-based simulation environment, a multi-agent deep reinforcement learning controller, a safety action projection mechanism, and an EV constraint handling mechanism.

2.1 Overall Framework

The overall workflow can be summarized as follows. First, the historical data are processed into time-series sequences of load demand, PV generation, and electricity prices. The LSTM predictor then generates forecasts over a future prediction horizon. The environment constructs the observation by combining the current system state with the predicted sequences. Based on its own observation, each MADRL actor outputs a control action. The generated actions subsequently undergo local feasibility checks and, if enabled, a safety action projection procedure. The environment then updates the battery SoC, EV SoC, and the net load of each prosumer according to the final control actions. Next, pandapower performs the power flow calculation using the updated net loads. Finally, the environment computes the reward based on the electricity cost, EV charging requirements, battery behavior, and the distribution network operating condition, and returns the reward to the agents for training.

Let x_t denote the external input data at the current time step, including load $L_{i,t}$, PV generation $G_{i,t}$, and electricity price p_t . Let $\hat{x}_{t:t+H-1}$ denote the prediction window of length H . The process within a time step is given by

$$x_t, \hat{x}_{t:t+H-1}, SOC_t^b, SOC_t^{ev} \rightarrow o_t \rightarrow a_t \rightarrow \bar{a}_t \rightarrow s_{t+1}, r_t.$$

where o_t is the observation, a_t is the actor's raw action, $\bar{a}_t$ is the executed action after feasibility checks or projection, s_{t+1} is the environment's next state, and r_t is the immediate reward for each agent.

The overall framework of this work includes the following steps:

1. Read historical household load, PV generation, and electricity price data.

2. Use LSTM model to generate future prediction sequences for electricity price, load, and PV generation.
3. The environment constructs the observation based on the current SoC, EV connection status, current exogenous variables, and the prediction window.
4. Each agent's actor outputs three action components: battery action, PV action, and EV action.
5. The battery action is first mapped to the feasible power range of SoC; in MADRL_PROJECTION, battery power and PV curtailment are further subjected to grid safety projection.
6. The environment updates the battery SoC, EV SoC, PV utilization, and net load based on the final actions.
7. pandapower writes the prosumer net load to the low-voltage distribution network and performs power flow calculation.
8. The environment calculates the reward based on economic terms, EV-related terms, battery-related terms, and grid safety terms.
9. The transition is written to the replay buffer and is used to update the actor and critic networks.

Figure 2.1 illustrates the main workflow of this work.

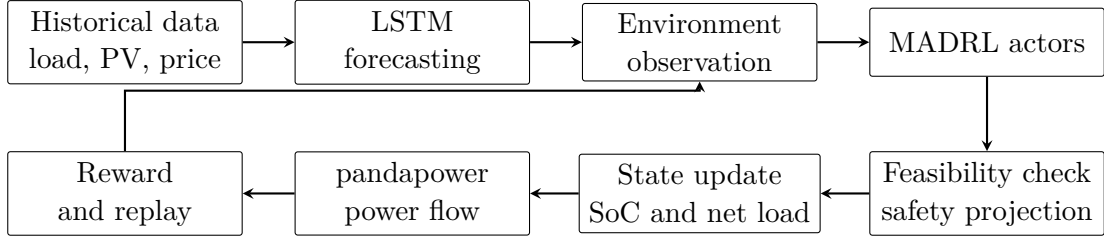


Figure 2.1: The work flow of the MADRL energy management framework.

2.2 LSTM Forecasting

This thesis employs an LSTM-based forecasting module to provide future-horizon information to the controller. The forecasting targets include the electricity price, prosumer load, and PV generation. The prediction results determine the amount of future

information available to the controller and therefore directly influence the scheduling performance of the stationary battery, PV curtailment, and EV charging.

For any scalar sequence y_t , LSTM uses a historical window of length h as input and outputs a predictive window of length H . The current project configuration is

$$h = 192, \quad H = 49.$$

Since the simulation time step is 0.25 h, a historical window of 192 time steps corresponds to the previous 48 hours, while a prediction window of 49 time steps includes the current time step and approximately the following 12 hours. The training sample can be formulated as

$$\begin{aligned} X_k &= (y_{k-h}, y_{k-h+1}, \dots, y_{k-1}), \\ Y_k &= (y_k, y_{k+1}, \dots, y_{k+H-1}). \end{aligned}$$

The LSTM learns the mapping

$$\hat{Y}_k = f_\theta(X_k, \phi_k),$$

where ϕ_k denotes the optional temporal features, such as the hour of the day, the day of the week, or annual cyclic features.

In this work, the electricity price is predicted using a separate LSTM model; the PV generation is predicted using a shared model trained on reference PV sequences; and the load prediction is organized by prosumer and load components.

The predicted sequences used in the environment are

$$\hat{p}_{t:t+H-1}, \quad \hat{L}_{i,t:t+H-1}, \quad \hat{G}_{i,t:t+H-1}.$$

In the perfect forecast mode, these sequences are directly obtained from the ground-truth future values. In the LSTM forecast mode, they are generated by the trained LSTM predictor. Unless otherwise specified, the MADRL controller is trained using the LSTM forecast mode.

2.3 MADRL Observation

Observation is divided into three components: the local observation, the forecast sequence observation, and the safety observation. This design follows the centralized training and decentralized execution (CTDE) paradigm. Specifically, each actor makes decentralized decisions primarily based on its own local state and the forecast horizon, while the critic has access to the joint information of all agents during the training process.

2.3.1 Local Observation

The local observation contains the key physical states of each agent at the current time step:

$$o_{i,t}^{local} = [SOC_{i,t}^b, SOC_{i,t}^{ev}, A_t^{ev}, L_{i,t}, G_{i,t}, p_t].$$

Here, $SOC_{i,t}^b$ denotes the SoC of the stationary battery, $SOC_{i,t}^{ev}$ denotes the EV SoC, A_t^{ev} indicates whether the EV is connected to the residential charger, $L_{i,t}$ represents the current electrical load, $G_{i,t}$ denotes the currently available PV generation, and p_t is the current electricity price.

In the MADRL actor, the local information is further transformed into `madr1_local`. This feature includes intraday and annual cyclic time features, normalized battery SoC, normalized EV SoC, and EV availability indicators:

$$o_{i,t}^{madr1} = [\sin(2\pi h_t/24), \cos(2\pi h_t/24), \sin(2\pi d_t/365.25), \cos(2\pi d_t/365.25), \bar{SOC}_{i,t}^b, \bar{SOC}_{i,t}^{ev}, A_t^{ev}].$$

This representation enables the actor not only to perceive the current states of the devices but also to identify the current temporal position within both the daily and annual cycles.

2.3.2 Forecast Sequence Observation

The forecast sequence observation describes the future trajectories of the electricity price, load demand, and PV generation. In the environment, the **sequence** can be expressed as

$$o_{i,t}^{seq} = [\hat{p}_{t:t+H-1}, \hat{L}_{i,t:t+H-1}, \hat{G}_{i,t:t+H-1}].$$

In practice, the MADRL actor uses normalized or relative representations of these forecast sequences:

$$\bar{p}_{t:t+H-1}, \quad \Delta\bar{p}_{t:t+H-1}, \quad \bar{L}_{i,t:t+H-1}, \quad \bar{G}_{i,t:t+H-1}.$$

where $\bar{p}$ denotes the relative electricity price within the prediction horizon, $\Delta\bar{p}$ represents the price variation intensity over the prediction window, and $\bar{L}$ and $\bar{G}$ denote the normalized load and PV generation forecasts, respectively. Physically, these features enable the actor to schedule the stationary battery and EV charging according to anticipated periods of high or low electricity prices, while also determining whether battery capacity should be reserved or utilized based on the forecasted PV generation.

2.3.3 Safety Observation

The safety observation is used for action feasibility checking and the safety projection procedure. It contains the physical parameters required to determine the device operating

limits and perform the grid safety projection:

$$o_{i,t}^{safe} = [SOC_{i,t}^b, SOC_{i,t}^{ev}, A_t^{ev}, L_{i,t}, G_{i,t}, C_i^b, P_i^{b,max}, C_i^{ev}, P_i^{ev,max}].$$

Here, the battery capacity, battery maximum charging and discharging power, EV battery capacity, and EV maximum charging power are not merely static configuration parameters; they also directly determine the physically feasible action space. For example, the battery SoC together with $P_i^{b,max}$ determines the maximum amount of energy that can be charged into or discharged from the battery at the current time step. Similarly, the EV connection status together with $P_i^{ev,max}$ determines whether the EV is currently available for charging and, if so, the maximum allowable charging power.

2.4 MADRL Action Design

Each agent's actor outputs a three-dimensional action vector:

$$a_{i,t} = [a_{i,t}^b, a_{i,t}^{pv}, a_{i,t}^{ev}] \in [-1, 1]^3.$$

The three action components correspond to the stationary battery, PV utilization, and EV charging, respectively.

2.4.1 Battery Action Mapping

The raw battery action generated by the actor is not executed directly as the physical battery power. Instead, it is first mapped to the feasible power range determined by the current battery SoC. Given the current SoC, the battery cannot be charged beyond SOC_{max}^b or discharged below SOC_{min}^b within a single time step. Therefore, the feasible battery power limits at the current time step are given by

$$P_{i,t}^{b,low} = -\min\left(P_i^{b,max}, \frac{(SOC_{i,t}^b - SOC_{min}^b)C_i^b\eta_b}{\Delta t}\right),$$

$$P_{i,t}^{b,up} = \min\left(P_i^{b,max}, \frac{(SOC_{max}^b - SOC_{i,t}^b)C_i^b}{\eta_b\Delta t}\right).$$

The raw action $a_{i,t}^b \in [-1, 1]$ is linearly mapped to this interval:

$$P_{i,t}^b = P_{i,t}^{b,low} + \frac{a_{i,t}^b + 1}{2}(P_{i,t}^{b,up} - P_{i,t}^{b,low}).$$

This ensures that even if the actor outputs a boundary value, the executed action does not directly cause the battery SoC to violate its limits.

2.4.2 PV Action Mapping

The PV action represents the PV utilization factor. It is defined as

$$u_{i,t}^{pv} = \min \left(1, \max \left(0, \frac{a_{i,t}^{pv} + 1}{2} \right) \right).$$

The corresponding utilized PV generation and curtailed PV power are given by

$$G_{i,t}^{use} = u_{i,t}^{pv} G_{i,t},$$

$$G_{i,t}^{curt} = G_{i,t} - G_{i,t}^{use}.$$

When $a_{i,t}^{pv} = 1$, the agent uses all available PV; when $a_{i,t}^{pv} = -1$, the PV is completely curtailed. The physical meaning of the PV action is to reduce reverse power flow or voltage violation risks through controllable curtailment.

2.4.3 EV Action Mapping

The EV action represents the utilization factor of the residential charger. It is defined as

$$u_{i,t}^{ev} = \min \left(1, \max \left(0, \frac{a_{i,t}^{ev} + 1}{2} \right) \right).$$

Then, EV charging power requested by the actor is

$$P_{i,t}^{ev,rl} = A_t^{ev} u_{i,t}^{ev} P_i^{ev,max}.$$

If the EV is not connected, $A_t^{ev} = 0$, the EV charging power is always zero regardless of the action produced by the actor. Subsequently, the environment determines the final executed charging power $P_{i,t}^{ev}$ by considering the EV SoC headroom as well as the hard projection or emergency charging mechanism, if enabled.

2.5 Reward Function Design

The reward function combines the economic objective, EV mobility requirements, battery behavior, and distribution network safety. For each agent i , the total reward consists of multiple components. Positive terms represent benefits or incentives, whereas negative terms represent costs or constraint violations.

2.5.1 EV Charging Cost

EV charging draws electrical energy from either the grid or local energy resources. Its economic cost is determined by the actual charging power and the electricity price. Let

$$\tilde{p}_t = p_t + \mu^{imp}$$

denote the electricity price. The basic EV charging cost is then given by

$$C_{i,t}^{ev} = w_{cost}^{ev} P_{i,t}^{ev} \tilde{p}_t \Delta t.$$

The corresponding reward term is

$$r_{i,t}^{ev,cost} = -C_{i,t}^{ev}.$$

EV charging during high-price periods incurs higher costs, while charging during low-price periods is more economical.

2.5.2 EV Costweight Modelling

Experimental results show that using only the EV charging cost as the reward encourages overly conservative EV charging behavior, even during periods of low electricity prices. The parameter w_{cost}^{ev} is therefore introduced to adjust the relative importance of the EV charging cost in the MADRL reward function. In the current implementation, the soft and hard modes use the configuration parameter `ev_charging_cost_weight`, whereas the emergency mode uses a fixed value of 1:

$$w_{cost}^{ev} = \begin{cases} \text{ev_charging_cost_weight}, & \text{mode} \in \{\text{soft}, \text{hard}\}, \\ 1, & \text{mode} = \text{emergency}. \end{cases}$$

When $w_{cost}^{ev} > 1$, the agent places greater emphasis on avoiding costly EV charging during periods of high electricity prices. When $w_{cost}^{ev} = 1$, the EV charging cost is calculated directly according to the actual electricity price.

2.5.3 EV Departure Penalty

The EV departure requirement represents a physical mobility constraint, requiring the EV to reach the target SoC before departure. In the reward function, the departure penalty converts this requirement into a learning signal under the soft mode. The SoC deficit at the departure time step is defined as

$$g_{i,t}^{dep} = \max(0, SOC_{req}^{ev} - SOC_{i,t+1}^{ev}), \quad t_d = t^{dep}.$$

The corresponding penalty is

$$L_{i,t}^{ev,dep} = w_{dep}^{ev} (g_{i,t}^{dep})^2.$$

This penalty is activated only at the departure time step. Physically, it indicates that if the EV has not reached the required SoC by the time of departure, the user's requirement has not been satisfied. Consequently, the reward function assigns a larger negative feedback to discourage such behavior.

2.5.4 EV Progress Penalty

Experimental results indicate that applying the departure penalty only at the departure time step is still insufficient to consistently achieve the required departure SoC. To alleviate this issue, an additional progress penalty can be enabled in the soft EV experiments. This penalty provides a linear reference trajectory from the arrival SoC to the required departure SoC throughout the EV connection window. Let $\alpha_t \in [0, 1]$ denote the normalized progress within the current connection window. Then,

$$SOC_t^{ev,prog} = SOC_{arr}^{ev} + \alpha_t (SOC_{req}^{ev} - SOC_{arr}^{ev}).$$

if the EV SoC falls below this reference trajectory, a penalty is applied to encourage the agent to gradually approach the departure target during the night connection window. The progress gap can be expressed as

$$g_{i,t}^{prog} = \max(0, SOC_t^{ev,prog} - SOC_{i,t+1}^{ev}),$$

The corresponding penalty is

$$L_{i,t}^{ev,prog} = w_{prog}^{ev} (g_{i,t}^{prog})^2.$$

This term encourages the EV to gradually approach the required departure SoC throughout the overnight connection window, rather than postponing the entire charging demand until the final time steps before departure.

2.5.5 Price-Aware Penalty

During the experiments, it was further observed that increasing the EV costweight introduces competition between the EV reward and the battery reward, leading to unsatisfactory overall performance. Moreover, the EV is still unable to completely avoid charging during periods of high electricity prices. To further encourage the EV to avoid expensive charging periods, a price-aware penalty is introduced in this thesis. First, a quantile threshold is computed from the electricity price profile within the current episode:

$$p^{thr} = Q_q(\{p_\tau\}),$$

where q denotes `ev_price_aware_threshold_quantile`, whose used value is 0.70. When the current electricity price exceeds this threshold, the normalized excess electricity price is defined as

$$\delta_t^p = \frac{\max(0, \tilde{p}_t - p^{thr})}{\max(p^{thr}, 10^{-6})}.$$

The price-aware penalty is then defined as

$$L_{i,t}^{ev,price} = w_{price}^{ev} P_{i,t}^{ev} \delta_t^p \Delta t.$$

When the electricity price does not exceed the threshold, this penalty is zero. When the electricity price exceeds the threshold and the EV is charging, this term becomes a positive penalty, encouraging the agent to shift flexible EV charging demand to periods with relatively lower electricity prices.

2.5.6 Battery Reward and Regularization

The battery reward consists of four components: the economic benefit of battery charging and discharging, the boundary action penalty, the SoC regularization term, and the throughput bonus applied during the early stage of training. The instantaneous economic benefit of the battery is given by

$$r_{i,t}^{bat} = [\max(-P_{i,t}^b, 0) - \max(P_{i,t}^b, 0)] \tilde{p}_t \Delta t.$$

When the battery is already close to its SoC limits but continues to move toward the corresponding boundary, an action boundary penalty is imposed:

$$L_{i,t}^{bat,act} = w_{act}^{bat} |P_{i,t}^b| I_{i,t}^{bd}.$$

where $I_{i,t}^{bd} = 1$ indicates that the battery is discharging near a low SoC limit or charging near a high SoC limit; otherwise, $I_{i,t}^{bd} = 0$. Additionally, the SoC regularization term encourages the battery to stay away from the hard limits:

$$L_{i,t}^{bat,soc} = w_{soc}^{bat} [\max(0, SOC_{low}^{soft} - SOC_{i,t}^b)^2 + \max(0, SOC_{i,t}^b - SOC_{high}^{soft})^2].$$

The throughput bonus for battery applied during the early stage of training is given by

$$r_{i,t}^{bat,thr} = w_t^{thr} |P_{i,t}^b| \Delta t.$$

This term gradually decays with training progress, aiming to prevent the actor from learning overly conservative strategies that avoid using the battery entirely in the early stages of training.

2.5.7 Grid Safety Penalty

The grid safety penalty is derived from the results of the pandapower power flow calculation and consists of the three categories introduced previously: voltage violations, line loading violations, and transformer loading violations, as mentioned before. Let ψ_t^V , ψ_t^{line} , and ψ_t^{tr} denote the raw violation measures for voltage, line loading, and transformer loading, respectively. The corresponding penalty terms are

$$L_{i,t}^V = Nw_V\psi_t^V\omega_{i,t}^V,$$

$$L_{i,t}^{line} = w_{line}\psi_t^{line},$$

$$L_{i,t}^{tr} = w_{tr}\psi_t^{tr}.$$

where $\omega_{i,t}^V$ denotes the voltage penalty weight based on the degree of voltage violation for each agent bus; line and transformer penalties are currently the same for all agents. This design ensures that agents not only focus on their own economic benefits but also perceive the impact of their actions on the safety of the distribution grid.

2.5.8 Total Reward

The total reward for each agent can be written as

$$r_{i,t} = r_{i,t}^{bat} - C_{i,t}^{ev} - L_{i,t}^{bat,act} - L_{i,t}^{bat,soc} - L_{i,t}^{ev,dep} - L_{i,t}^{ev,proj} - L_{i,t}^{ev,price} - L_{i,t}^{ev,proj} - L_{i,t}^{ev,eme} + r_{i,t}^{bat,thr} - L_{i,t}^V - L_{i,t}^{line} - L_{i,t}^{tr}.$$

Here, $L_{i,t}^{ev,proj}$ and $L_{i,t}^{ev,eme}$ denote the diagnostic penalty terms introduced by the hard projection and emergency charging mechanisms, respectively, which will be discussed in the following sections. Overall, the reward function is designed to simultaneously encourage cost-efficient operation, satisfy the EV mobility requirements, promote appropriate battery utilization, and prevent violations of the distribution network safety constraints.

2.6 Safety Projection for Grid Constraints

The reward penalty represents a soft constraint: it allows the agent to take unsafe actions but penalizes such behavior through the reward function. In contrast, the safety projection acts as a hard correction mechanism applied before action execution. It directly modifies the control actions before they are executed by the environment, thereby moving them closer to the grid-safe feasible region. In the proposed MADRL_PROJECTION framework, the safety projection is employed as a complement to the reward penalty mechanism.

The decision variables of the projection are not the complete action vector but only the battery power and the PV curtailment:

$$x_t = [P_{1,t}^b, \dots, P_{N,t}^b, G_{1,t}^{curt}, \dots, G_{N,t}^{curt}].$$

Given the current load, PV generation, and EV charging power, the base net load is

$$P_{i,t}^{base} = L_{i,t} - G_{i,t} + P_{i,t}^{ev}.$$

The safety projector uses pandapower to compute finite-difference sensitivities of the bus voltages and line loading with respect to the agents' net power injections around the zero-injection operating point. The corresponding linearized relationships can be written as

$$\begin{aligned} V_t &\approx V^{base} + S^V \Delta P_t, \\ \lambda_t^{line} &\approx \lambda^{base} + S^{line} \Delta P_t. \end{aligned}$$

where ΔP_t is determined by the battery power and PV curtailment. The projection goal is to satisfy the linearized safety constraints with respect to voltage, line, and transformer limits while making minimal changes to the original actions.

Each linearized constraint can be written in half-space form

$$r_j^T x \leq b_j.$$

If the current solution x violates the corresponding constraint, it is projected sequentially onto each constraint hyperplane as

$$x \leftarrow x - \frac{\max(0, r_j^T x - b_j)}{\|r_j\|_2^2} r_j.$$

After the projection step, the battery power is clipped again to the locally feasible SoC range, while the PV curtailment is clipped to

$$0 \leq G_{i,t}^{curt} \leq G_{i,t}.$$

This procedure is repeated for several iterations. Physically, the projection mitigates voltage and line loading violations by reducing battery charging, increasing battery discharging, or increasing PV curtailment. Since EV charging is primarily driven by the user's mobility requirements, this thesis treats the EV constraints separately from the grid safety projection.

2.7 EV Departure Requirement Handling

This thesis implements three approaches for handling the EV departure requirement: the soft, hard, and emergency charging strategies. All three approaches employ the same physical EV model, but they differ in terms of action execution and reward feedback.

2.7.1 Soft Mode

Soft mode directly executes the EV charging power requested by the actor, and guides policy learning through departure penalties, progress penalties, or price-aware penalties in the reward function. The executed power is

$$P_{i,t}^{ev} = P_{i,t}^{ev,rl}$$

and is subject to constraints from EV connection, charger power, and SoC headroom. The advantage of soft mode is that it allows for high policy freedom, enabling agents to learn price responsiveness; however, its disadvantage is that the departure requirement mainly relies on reward learning, and if the reward signal is sparse, it may result in insufficient SoC at home.

2.7.2 Hard Mode

In the hard mode, the execution layer checks whether the current EV charging action is sufficient to preserve the feasibility of meeting the required departure SoC. Let K_t^{future} denote the number of remaining time steps during which the EV will remain connected before departure. The maximum achievable increase in SoC over the remaining connection window is

$$\Delta SOC_{i,t}^{future,max} = \frac{K_t^{future} P_i^{ev,max} \Delta t \eta_{ev}}{C_i^{ev}}.$$

If the EV is not charged sufficiently at the current time step, it may become impossible to reach SOC_{req}^{ev} before departure, even if the charger subsequently operates at its maximum power. Therefore, the hard mode computes the minimum required charging power at the current time step as

$$P_{i,t}^{ev,req} = \max \left(0, \frac{(SOC_{req}^{ev} - \Delta SOC_{i,t}^{future,max} - SOC_{i,t}^{ev}) C_i^{ev}}{\Delta t \eta_{ev}} \right).$$

The finally executed charging power is then determined by

$$P_{i,t}^{ev} = \min \left(P_i^{ev,max}, \max(P_{i,t}^{ev,rl}, P_{i,t}^{ev,req}) \right).$$

The physical meaning is that the control policy is free to select a charging power higher than the minimum required value, but it is not allowed to choose a charging power below the minimum necessary level.

2.7.3 Emergency Charging

Emergency charging is a remedial mechanism designed to compensate for the potential SoC deficiency that may occur under the soft mode. It is activated only within an emergency window immediately before the EV departure time. Let the emergency window consist of K^{eme} time steps. When the EV enters the emergency window and its SoC remains below the required target, the system computes the required charging power based on the remaining SoC deficit:

$$E_{i,t}^{need} = \max(0, SOC_{req}^{ev} - SOC_{i,t}^{ev})C_i^{ev},$$

$$P_{i,t}^{ev,eme} = \min\left(P_i^{ev,max}, \frac{E_{i,t}^{need}}{K_t^{rem}\Delta t\eta_{ev}}\right).$$

The finally executed power is then determined by

$$P_{i,t}^{ev} = \max(P_{i,t}^{ev,rl}, P_{i,t}^{ev,eme}).$$

Unlike the hard mode, the emergency mode does not continuously evaluate future feasibility throughout the entire EV connection window. Instead, it performs remedial charging only when the departure time is approaching. Therefore, the emergency mode is simpler to implement, but it may result in a large concentration of charging power during the final stage before departure.

2.8 MADRL Training Algorithm

This thesis adopts a multi-agent actor-critic framework based on the centralized training and decentralized execution paradigm. Each agent is equipped with an independent actor:

$$a_{i,t} = \pi_i(o_{i,t}; \theta_i),$$

where $o_{i,t}$ denotes the actor observation of agent i . During training, a centralized critic is employed, which receives the observations and the actions of all agents:

$$Q_i(o_{1:N,t}, a_{1:N,t}; \phi_i).$$

To reduce Q-value overestimation, the code employs twin critics for each agent:

$$Q_{i,1}, \quad Q_{i,2}.$$

Replay buffer saves transition:

$$(o_t, a_t, R_t^{(n)}, o_{t+n}, d_t),$$

where $R_t^{(n)}$ is the n-step return:

$$R_{i,t}^{(n)} = \sum_{k=0}^{n-1} \gamma^k r_{i,t+k}.$$

Here, $n = 48$ corresponds to an accumulated reward over approximately 12 hours. The critic target is given by

$$y_{i,t} = R_{i,t}^{(n)} + \gamma^n (1 - d_t) \min(Q'_{i,1}(o_{t+n}, a'_{t+n}), Q'_{i,2}(o_{t+n}, a'_{t+n})).$$

Target actions are added with TD3-style target policy smoothing noise:

$$a'_{t+n} = \text{guard}(\pi'(o_{t+n}) + \epsilon).$$

where **guard** denotes the local SoC feasibility check and optional safety projection.

Critic loss is

$$L_i^Q = (Q_{i,1}(o_t, a_t) - y_{i,t})^2 + (Q_{i,2}(o_t, a_t) - y_{i,t})^2.$$

During the actor update, only the i -th action in the joint action vector is replaced:

$$a_{i,t}^{pol} = \text{guard}(\pi_i(o_{i,t})),$$

$$L_i^\pi = -Q_{i,1}(o_t, a_{1,t}, \dots, a_{i,t}^{pol}, \dots, a_{N,t}).$$

Target network 采用软更新:

$$\theta'_i \leftarrow (1 - \tau)\theta'_i + \tau\theta_i,$$

$$\phi'_i \leftarrow (1 - \tau)\phi'_i + \tau\phi_i.$$

During training, both exploration noise and feasible random exploration are employed. The former adds random perturbations to the actor output, while the latter samples locally feasible battery actions with a certain probability during the early stage of training, thereby helping the agent explore the feasible battery SoC operating region.

2.8.1 Training Pseudocode

The overall MADRL training procedure adopted in this thesis can be summarized by the following pseudocode:

1. Initialize N actors, N twin critics, the corresponding target networks, and the replay buffer.

2. Reset the parallel training environments and obtain the initial observations.
3. For each environment step:
 - a) Each actor generates a raw action based on its current observation.
 - b) Add exploration noise and perform the local feasibility check.
 - c) If MADRL_PROJECTION is enabled, perform the safety projection.
 - d) Execute the actions in the environment, update the battery SoC, EV SoC, and net load, and perform the power flow calculation using pandapower.
 - e) The environment returns the reward, the next observation, and the done flag.
 - f) Store the transition in the replay buffer and compute the n -step return.
 - g) Once the replay buffer contains sufficient samples, draw a mini-batch to update the critics.
 - h) Update the actors and target networks every predefined number of training steps.
4. Repeat the above procedure until the specified number of training episodes is reached.

The algorithm is decentralized during execution: each actor only needs its own local observation and prediction sequence to output an action; however, during training, the critic uses joint observations and joint actions to learn the coupling effects between agents through the power grid flow.

2.9 Baseline Controllers

To evaluate the performance of the proposed MADRL approach, this thesis also implements optimization-based baseline controllers, including Local MPC and ADMM-MPC.

2.9.1 Local MPC

Local MPC solves an independent receding-horizon optimization problem for each prosumer. It neither considers the actions of the other prosumers nor explicitly enforces network-wide voltage or line loading constraints. Instead, it optimizes its own battery operation, PV curtailment, and EV charging schedule over the prediction horizon. For prosumer i , the primary decision variables of the Local MPC formulation include

$$c_{i,\tau}, d_{i,\tau}, G_{i,\tau}^{curt}, P_{i,\tau}^{ev}, E_{i,\tau}^b, E_{i,\tau}^{ev}.$$

where c and d denote the battery charging power and discharging power, respectively, with

$$P_{i,\tau}^b = c_{i,\tau} - d_{i,\tau}.$$

Function for the battery is:

$$E_{i,\tau+1}^b = E_{i,\tau}^b + \eta_b c_{i,\tau} \Delta t - \frac{d_{i,\tau} \Delta t}{\eta_b}.$$

Function for the EV is:

$$E_{i,\tau+1}^{ev} = E_{i,\tau}^{ev} + \eta_{ev} P_{i,\tau}^{ev} \Delta t.$$

Local net load is:

$$P_{i,\tau}^{net} = L_{i,\tau} - G_{i,\tau} + G_{i,\tau}^{curt} + c_{i,\tau} - d_{i,\tau} + P_{i,\tau}^{ev}.$$

Local MPC objective function mainly minimizes the battery charging cost, discharging benefit, and EV charging cost:

$$\min_{\tau} \sum_{\tau} \tilde{p}_{\tau} \Delta t (c_{i,\tau} - d_{i,\tau} + P_{i,\tau}^{ev}) + \epsilon_b \Delta t (c_{i,\tau} + d_{i,\tau}) + w_{dep}^{ev} g_i^2.$$

where ϵ_b is a small throughput tie-breaker, used to avoid unnecessary simultaneous charging and discharging or equivalent solutions; g_i is the EV departure gap. The advantage of Local MPC is that it has a small problem size and fast solution time; its disadvantage is the lack of grid coordination across agents.

2.9.2 ADMM-MPC

ADMM-MPC introduces distributed coordination on top of Local MPC, aiming to constrain the impact of aggregated net load on transformers. Each agent still solves its local subproblem, but additionally introduces local net load contribution

$$q_{i,\tau} = \alpha_{i,\tau} P_{i,\tau}^{net}.$$

In the current implementation, $\alpha_{i,\tau} = 1$. The global coordination objective is to ensure that all agents' contributions satisfy the transformer aggregation bounds:

$$-\bar{P}^{tr} \leq \sum_i q_{i,\tau} \leq \bar{P}^{tr}.$$

ADMM adds a quadratic penalty term to the local objective function :

$$J_i^{ADMM} = J_i^{local} + \frac{\rho}{2} \|\alpha_i P_i^{net} - z_i + u_i\|_2^2.$$

where z_i is the coordination variable, u_i is the scaled dual variable, and ρ is the ADMM penalty parameter. The solution process iterates between local optimization and aggregation projection:

$$\begin{aligned} z &\leftarrow \Pi_{[-p_{tr}, p_{tr}]}(q + u), \\ u &\leftarrow u + q - z. \end{aligned}$$

ADMM-MPC preserves the local optimization structure of each agent while reducing the risk of transformer overloading through coordination based on coupling constraints. Compared with Local MPC, it provides a certain degree of system-level coordination. Compared with a centralized MISOCP formulation, its computational structure is more decentralized.

3 Experimental Setup

This chapter describes the data, device parameters, training parameters, grid configuration, and evaluation metrics used in the experiments.

3.1 Dataset and Simulation Period

The data used in this chapter is located in the project directory `datasets/prosumer/`. The data includes household load, PV reference generation, and electricity price. The load profiles for the three prosumers are

SFH12, SFH18, SFH20.

Training and evaluation use different years and date ranges. The current configuration is shown in Table 3.1.

Table 3.1: Dataset and Simulation Time Period.

Item	Setting	Description
Data Directory	<code>datasets/prosumer</code>	Load, PV, and electricity price data
Training Year	2019	Used for LSTM and MADRL training
Training Dates	2019-01-01 to 2019-12-31	Main training data period
Evaluation Year	2020	Used for controller evaluation
Evaluation Dates	2020-04-01 to 2020-04-15	Main evaluation data period
Time Step	0.25 h	15-minute resolution
Daily Steps	96	One natural day episode
Load Profiles	SFH12, SFH18, SFH20	Three prosumers

During the evaluation phase, 15 evaluation episodes are selected by default. Each episode corresponds to the operation process of one day, where the controller reads observations and outputs actions at each time step, and the environment updates the system state.

3.2 Battery and EV Parameters

Battery parameters are shown in Table 3.2.

Table 3.2: Stationary battery parameters.

Parameter	Value	Description
C_i^b	100 kWh	Battery capacity for each prosumer
$SOC_{\min}^b$	0.05	Minimum battery SoC
$SOC_{\max}^b$	0.95	Maximum battery SoC
SOC_0^b	0.05	Initial battery SoC for evaluation
η_b	0.95	Battery charging/discharging efficiency
$r_{\max}^b$	0.5	Maximum charging/discharging rate
$P_i^{b,\max}$	50 kW	Maximum charging/discharging power

The maximum power of the battery is given by its capacity and maximum charging rate:

$$P_i^{b,\max} = r_{\max}^b C_i^b = 0.5 \times 100 = 50 \text{ kW}.$$

This setting allows the battery to charge when prices are low or PV generation is sufficient, and discharge when prices are high or load is significant, while keeping the SoC within the bounds $[0.05, 0.95]$.

EV parameters are shown in Table 3.3.

Table 3.3: EV parameters.

Parameter	Value	Description
C_i^{ev}	60 kWh	Battery capacity for each EV
$SOC_{\min}^{ev}$	0.10	Minimum EV SoC
$SOC_{\max}^{ev}$	0.95	Maximum EV SoC
SOC_{arr}^{ev}	0.30	SoC at home arrival
SOC_{req}^{ev}	0.90	Required SoC at departure
$P_i^{ev,\max}$	11 kW	Maximum charging power for home charging
η_{ev}	0.95	EV charging efficiency
t^{arr}	72	Home arrival time (18:00)
t^{dep}	28	Home departure time (07:00)

3.3 Training Parameters

3.3.1 LSTM Training Parameters

The LSTM forecasting module predicts the electricity price, load demand, and PV generation. The historical input window is set to

$$h = 192,$$

corresponding to the previous 48 hours, while the prediction horizon is

$$H = 49,$$

corresponding to the current time step and approximately the following 12 hours. The LSTM training parameters are summarized in Table 3.4.

Table 3.4: LSTM forecasting model training parameters.

Parameter	Value
history window	192
sequence length	49
batch size	1024
train ratio	0.70
validation ratio	0.15
price epochs	50
price hidden size	128
price layers	2
load epochs	20
load hidden size	64
PV epochs	20
PV hidden size	64

3.3.2 MADRL Training Parameters

The training parameters of the MADRL framework are summarized in Table 3.5.

The MADRL approaches compared in the experiments include `MADRL_BASE`, which does not incorporate any safety mechanism, and `MADRL_PROJECTION`, which applies a safety projection before action execution.

Table 3.5: MADRL training parameters.

Parameter	Value
action dimension	3
hidden dimension	256
discount factor γ	0.999
soft update τ	0.01
policy update frequency	2
policy noise	0.2
noise clip	0.5
train episodes	20
number of environments	1
replay buffer size	50000
batch size	128
actor learning rate	10^{-4}
critic learning rate	10^{-4}
n-step return	48
exploration noise	0.4 to 0.2

3.4 Network Configuration

The network security limits adopted in the experiments are summarized in Table 3.6.

3.5 Evaluation Metrics

Evaluation metrics can be divided into four categories: economic metrics, grid safety metrics, EV departure SoC metrics, and diagnostic metrics for the relationship between EV/battery and electricity prices. These metrics are used to evaluate the controller’s performance from different perspectives, but this chapter only defines the metrics without discussing specific results.

3.5.1 Economic Metrics

The economic metrics are used to evaluate the operating cost of the controller and the economic benefits achieved through energy storage. The main metrics include:

$$C_{total}^{bat} = \sum_{t,i} \max(P_{i,t}^b, 0) \tilde{p}_t \Delta t - \sum_{t,i} \max(-P_{i,t}^b, 0) \tilde{p}_t \Delta t,$$

Table 3.6: Network configuration and security limits.

Parameter	Value
SimBench code	1-LV-rural1--0-sw
agent bus ids	12, 4, 2
power flow solver	Newton-Raphson
$V_{\min}$	0.95 p.u.
$V_{\max}$	1.05 p.u.
line loading limit	100%
transformer loading limit	100%

$$C_{total}^{ev} = \sum_{t,i} P_{i,t}^{ev} \tilde{p}_t \Delta t,$$

$$C^{total} = C_{total}^{bat} + C_{total}^{ev} + C_{total}^{pv}.$$

Besides the total cost, the records also include storage charge cost, storage discharge revenue, storage profit, and episode reward.

3.5.2 Grid Safety Metrics

Grid safety metrics are derived from the pandapower power flow results. They mainly include:

$$\begin{aligned} & \min V_{b,t}, \quad \max V_{b,t}, \\ & \text{voltage violation steps,} \\ & \text{trafo loading max pct,} \quad \text{trafo overload steps,} \\ & \text{feeder netload ramp mean abs.} \end{aligned}$$

where voltage violation steps represents the number of time steps where at least one bus voltage violates the bounds; trafo loading max pct represents the maximum transformer loading during the evaluation; feeder netload ramp is used to measure the severity of changes in the feeder's net load.

3.5.3 EV Departure SoC Metrics

EV metrics are used to evaluate whether users' travel demands are met. The core metrics include departure SoC, departure gap, EV session energy, and EV session cost. The departure SoC gap is defined as

$$g_i^{dep} = \max(0, SOC_{req}^{ev} - SOC_{i,t^{dep}}^{ev}).$$

If $g_i^{dep} = 0$, it indicates that agent i meets the target SoC when departing; if $g_i^{dep} > 0$, it indicates that the departure demand is not fully satisfied. The evaluation also records the total charging amount for each EV session:

$$E_i^{ev,session} = \sum_{t \in \mathcal{T}^{ev}} P_{i,t}^{ev} \Delta t,$$

3.5.4 Relationship between EV/Battery and Electricity Prices

To analyze whether the controller has learned price-aware scheduling, this paper also records the time-aligned sequences of EV charging power, battery charging/discharging power, and import price.

$$P_{i,t}^{ev} \text{ vs. } \tilde{p}_t,$$

$$P_{i,t}^b \text{ vs. } \tilde{p}_t.$$

A controller with good price responsiveness should schedule EV charging and battery charging predominantly during periods of low electricity prices, while battery discharging should occur primarily during periods of high electricity prices.

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